



Saint Columban College
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Safe Opener

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Medium

Reverse Engineering

picoCTF 2022

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Description

Can you open this safe?

I forgot the key to my safe but this [program](#) is supposed to help me with retrieving the lost key. Can you help me unlock my safe?

Put the password you recover into the picoCTF flag format like:

`picoCTF{password}`

Hints ?

(None)

21,528 users solved



91% Liked



`picoCTF{pl3as3_l3t_m3_1nt0_th3_saf3}`

Submit Flag

Author: The Analyst: Hyposelenia

Challenge: Safe Opener

Category: Reverse Engineering

Date: 4/10/26



I. Objective

The objective of this challenge is to **analyze a Java program**, identify an encoded string inside it, decode that string, and retrieve the hidden flag.

II. Background

This challenge involves **reverse engineering**, which means:

Looking at a program's code to understand what it is doing

The program is written in **Java**.

Important Concepts:

1. Hardcoded Values

Sometimes, programs store important data directly inside the code:

```
String encodedKey = "SGVsbG8=";
```

This is called a **hardcoded value**

2. Base64 Encoding

The string we saw is encoded using **Base64**.

Base64:

- Converts readable text into a different format
- Often used to safely store or transfer data
- Looks like random letters, numbers, +, /, =

Example:

- SGVsbG8= → decodes to Hello

Important: Base64 is **not encryption**, just encoding (easy to reverse)

III. Tool Used

- **VS Code** - used to open and read the Java source code



- **CyberChef** - used to decode the Base64 encoded string

IV. Methodology

1. Downloaded the provided file from the challenge

is program is
you help me u

2. Opened the Java source code using VS Code
3. Carefully read through the code

```
Downloads > % - safeOpener.java > Language Support for Java(TM) by Red Hat > safeOpener > opensafe(String,  
public class SafeOpener {  
    public static void main(String args[]) throws IOException {  
        break;  
    }  
}  
  
public static boolean openSafe(String password) {  
    String encodedkey = "cGwzYXMzX2wzdF9tM18xbnQwX3RoM19zYWYz";  
  
    if (password.equals(encodedkey)) {  
        System.out.println(x: "Sesame open");  
        return true;  
    }  
    else {  
        System.out.println(x: "Password is incorrect\n");  
        return false;  
    }  
}
```

4. Found a variable that stores an encoded string:

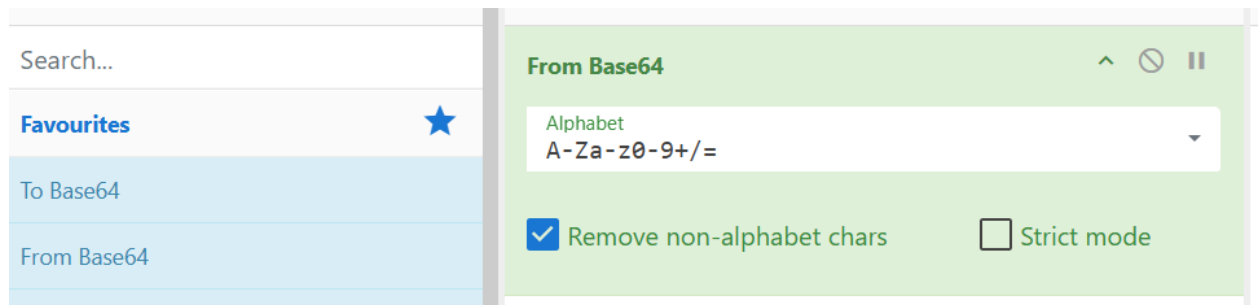
String encodedKey = "...";

```
public static boolean openSafe(String password) {  
    String encodedkey = "cGwzYXMzX2wzdF9tM18xbnQwX3RoM19zYWYz";  
    if (password.equals(encodedkey)) {
```

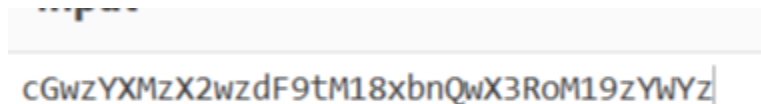
5. Recognized that the string is **Base64 encoded**
6. Copied the encoded string
7. Opened CyberChef



8. Used the “From Base64” operation



9. Pasted the encoded string into CyberChef

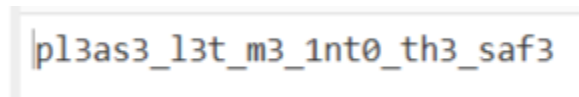


10. Decoded the string

11. The decoded result revealed the flag

V. Result

picoCTF{pl3as3_l3t_m3_1nt0_th3_saf3}



VI. Explanation

Step 1: The Program Hides Something

Inside the Java code, we see:

```
String encodedKey = "...";
```

This is suspicious because:

- It looks like random text
- It is stored directly in the code

Step 2: Recognizing Base64



The string:

- Ends with =
- Uses only certain characters

These are strong signs of **Base64 encoding**

Step 3: Decoding It

We used CyberChef to decode it:

- Input → encoded string
- Operation → “From Base64”
- Output → readable text

Step 4: The Result

After decoding, we get:

picoCTF{pl3as3_l3t_m3_1nt0_th3_saf3}

That is the flag

Simple Analogy

Think of it like a **locked safe**:

- The encoded string = locked message
- Base64 = simple lock
- CyberChef = key

Once we use the key, the message opens instantly.

VII. Conclusion

This challenge shows that:

- Reverse engineering often starts by **reading the source code carefully**
- Important data can be hidden in **plain sight** using encoding
- Recognizing patterns like Base64 is a valuable skill

By identifying and decoding the encoded string, we successfully retrieved the flag.

— **The Analyst: Hyposelenia**